Notes – WIP with Corrected CI Derivation

July 20, 2026

1 Introduction

This document summarizes what we have achieved so far with our topological model for collective conical intersection points.

The basic arrangement of our system is that three identical diatomic molecules are placed in a cavity in a Cartesian coordinate system, where the axes of the molecules are parallel to the x, y, and z axes. This arrangement can fundamentally create the formation of CIs in the system.

2 Model Potentials

Our system is described by two potential energy surfaces:

$$V_x(x) = x^2 \tag{1}$$

$$V_a(x) = a \cdot (x - x_{\text{shift}})^2 + c_0 \tag{2}$$

Where:

- a is a scaling parameter
- x_{shift} is a shift parameter for the second potential
- c_0 is a constant energy offset

3 Finding Conical Intersection Points

A conical intersection occurs when the two potential energy surfaces meet at a point. For our model, this happens when:

$$V_a(x) - V_x(x) = 0 \tag{3}$$

3.1 Mathematical Derivation

Starting with the difference between the potentials:

$$V_a(x) - V_x(x) = a \cdot (x - x_{\text{shift}})^2 + c_0 - x^2 \tag{4}$$

$$= (a-1) \cdot x^2 - 2 \cdot a \cdot x_{\text{shift}} \cdot x + c_1 \tag{5}$$

Where $c_1 = a \cdot x_{\text{shift}}^2 + c_0$.

We can rewrite this in the form of a perfect square plus a constant:

$$V_a(x) - V_x(x) = (a-1) \cdot (x - x_{\text{prime}})^2 + c_1' \tag{6}$$

Where x_{prime} is the point where the quadratic term has its minimum:

$$x_{\text{prime}} = \frac{a}{a-1} \cdot x_{\text{shift}} \tag{7}$$

4 Geometric Configuration of CI Points

For a 3D system with coordinates (r_1, r_2, r_3) representing the interatomic distances of three molecules, we need to find points where the potentials are equal. This leads to a specific geometric arrangement.

4.1 Trivial and Nontrivial CIs

4.1.1 Trivial CIs

Trivial CIs occur when all three interatomic distances are equal:

$$(r_1, r_2, r_3) \text{ with } r_1 = r_2 = r_3 (= r_0) \tag{8}$$

These form a seam with the property $r_1 = r_2 = r_3$.

4.1.2 Nontrivial CIs

To find nontrivial CIs, we search for points where two different interatomic distances $r_1 \neq r_2$ give equal potential differences:

$$(V_a - V_x)(r_1) = (V_a - V_x)(r_2) \tag{9}$$

If such $r_1 \neq r_2$ exist, the following configurations provide nontrivial CIs:

$$\text{CI}_1 : (r_1, r_2, r_3) = (r_1, r_1, r_2) \tag{10}$$

$$\text{CI}_2 : (r_1, r_2, r_3) = (r_1, r_2, r_1) \tag{11}$$

$$\text{CI}_3 : (r_1, r_2, r_3) = (r_2, r_1, r_1) \tag{12}$$

$$\text{CI}_4 : (r_1, r_2, r_3) = (r_2, r_2, r_1) \tag{13}$$

$$\text{CI}_5 : (r_1, r_2, r_3) = (r_2, r_1, r_2) \tag{14}$$

$$\text{CI}_6 : (r_1, r_2, r_3) = (r_1, r_2, r_2) \tag{15}$$

4.2 Geometric Constraints

Since the trivial CIs form a seam, we set up our closed paths in the plane perpendicular to this "trivial" seam. All points in this plane have the property:

$$r_1 + r_2 + r_3 = 3 \cdot r_0 \tag{16}$$

where (r_0, r_0, r_0) is the trivial CI in the plane.

It is easy to see that CI_1, CI_2, and CI_3 lie in the same plane, and similarly, CI_4, CI_5, and CI_6 also lie in the same plane.

4.3 Derivation for CI_1, CI_2, CI_3

We now search for nontrivial CIs of type CI_1, CI_2, CI_3 with the additional requirement:

$$r_1 + r_1 + r_2 = 3 \cdot r_0 \tag{17}$$

for a predefined r_0.

From the equality condition $(V_a - V_x)(r_1) = (V_a - V_x)(r_2)$ with $r_1 \neq r_2$, we obtain:

$$r_1 = x_{\text{prime}} - \delta, \quad r_2 = x_{\text{prime}} + \delta \tag{18}$$

where δ is an arbitrary parameter.

Applying the geometric constraint:

$$\begin{aligned} r_1 + r_1 + r_2 &= 3 \cdot r_0 && (19)\\ 3 \cdot x_{\text{prime}} - \delta &= 3 \cdot r_0 && (20)\\ \delta &= 3 \cdot (x_{\text{prime}} - r_0) && (21) \end{aligned}$$

Substituting back to find r_1 and r_2:

$$\begin{aligned} r_1 &= x_{\text{prime}} - \delta = 3 \cdot r_0 - 2 \cdot x_{\text{prime}} = r_0 - 2 \cdot (x_{\text{prime}} - r_0) && (22)\\ r_2 &= x_{\text{prime}} + \delta = 4 \cdot x_{\text{prime}} - 3 \cdot r_0 = r_0 + 4 \cdot (x_{\text{prime}} - r_0) && (23) \end{aligned}$$

4.4 Distance from Trivial CI

To determine how many CIs are enclosed by a given circle, we need to know the distance between the nontrivial and trivial CIs. The distance in the 3D coordinate space is:

$$\begin{aligned} d_{\text{CI}} &= |r_2 - r_0| \cdot \frac{\sqrt{6}}{2} && (24)\\ &= 4 \cdot |x_{\text{prime}} - r_0| \cdot \frac{\sqrt{6}}{2} && (25)\\ &= 2 \cdot \sqrt{6} \cdot |x_{\text{prime}} - r_0| && (26) \end{aligned}$$

4.5 Analysis of CI_4, CI_5, CI_6

Similar expressions could be derived for CI_4, CI_5, and CI_6. In this case, we would obtain:

$$\delta' = -3 \cdot (x_{\text{prime}} - r_0) = -\delta \tag{27}$$

Taking this value and substituting into the expressions for r_1 and r_2:

$$r_1' = x_{\text{prime}} - \delta' = x_{\text{prime}} + \delta = r_2 \tag{28}$$

$$r_2' = x_{\text{prime}} + \delta' = x_{\text{prime}} - \delta = r_1 \tag{29}$$

Combining this with the definitions of the nontrivial CI_i configurations, we realize that these CIs are actually the same as CI_1, CI_2, and CI_3. More specifically:

$$\mathrm{CI}_4 = \mathrm{CI}_1 \tag{30}$$

$$\mathrm{CI}_5 = \mathrm{CI}_2 \tag{31}$$

$$\mathrm{CI}_6 = \mathrm{CI}_3 \tag{32}$$

5 Implementation in Our Code

5.1 Circle Parameterization

In our code, we set up a perfect circle in the 3D configuration space that is orthogonal to the $(1, 1, 1)$ line. The center of this circle is at the origin $R_0 = (0, 0, 0)$.

The circle is parameterized by the angle θ with the following parameters:

- Circle radius: $d = 0.001$
- θ range: $[0, 2\pi]$
- Number of points: 5001
- Center point: $R_0 = (0, 0, 0)$

The parameter vector $\boldsymbol{R}_\theta$ traces this circle, which lies in the plane perpendicular to the $r_1 = r_2 = r_3$ line (the trivial CI seam).

5.2 Trivial CI Location

With our current potential parameters ($a_{V_x} = 1.0$, $a_{V_a} = 1.3$, $x_{\text{shift}} = 0.1$, $c_0 = 0.01$), the trivial CI is located at:

$$x_{\text{prime}} = \frac{a_{V_a}/a_{V_x}}{a_{V_a}/a_{V_x} - 1} \cdot x_{\text{shift}} = \frac{1.3}{1.3 - 1} \cdot 0.1 = 0.433 \tag{33}$$

Thus, the trivial CI is at $R_{\text{trivial}} = (0.433, 0.433, 0.433)$.

5.3 Non-trivial CI Locations

Using our circle center at $R_0 = (0, 0, 0)$ as the reference point ($r_0 = 0$), we can calculate the non-trivial CI points:

$$r_0 = 0 \tag{34}$$
$$r_1 = 3r_0 - 2x_{\text{prime}} = 3 \cdot 0 - 2 \cdot 0.433 = -0.866 \tag{35}$$
$$r_2 = 4x_{\text{prime}} - 3r_0 = 4 \cdot 0.433 - 3 \cdot 0 = 1.732 \tag{36}$$

The three distinct non-trivial CI configurations are:

$$\text{CI}_1 : (r_1, r_1, r_2) = (-0.866, -0.866, 1.732) \tag{37}$$
$$\text{CI}_2 : (r_1, r_2, r_1) = (-0.866, 1.732, -0.866) \tag{38}$$
$$\text{CI}_3 : (r_2, r_1, r_1) = (1.732, -0.866, -0.866) \tag{39}$$

5.4 Distance from Circle Center to CIs

The distance from our circle center $(0, 0, 0)$ to the non-trivial CIs is:

$$d_{\text{CI}} = 2 \cdot \sqrt{6} \cdot |x_{\text{prime}} - r_0| \tag{40}$$
$$= 2 \cdot \sqrt{6} \cdot |0.433 - 0| \tag{41}$$
$$= 2 \cdot \sqrt{6} \cdot 0.433 \tag{42}$$
$$\approx 2.121 \tag{43}$$

Since our circle radius is $d = 0.001$, which is much smaller than $d_{\text{CI}} \approx 2.121$, the circle does not enclose any of the non-trivial CI points.

5.5 Code Implementation

The function `find_nontrivial_CIs(aVx, aVa, x_shift, r0=None)` is used to calculate these CI points. When called with $r_0 = 0$, it returns:

- `trivial_CI`: $(0.433, 0.433, 0.433)$ - the trivial CI
- `nontrivial_CIs`: $[(-0.866, -0.866, 1.732), (-0.866, 1.732, -0.866), (1.732, -0.866, -0.866)]$
- `r0`: 0
- `r1`: -0.866
- `r2`: 1.732
- `x_prime`: 0.433
- `d_CI`: 2.121

5.6 Arrowhead Hamiltonian

Since the cavity excites collectively, the system is described by a 4×4 arrowhead Hamiltonian matrix with the following structure:

$$H(\boldsymbol{R}_\theta) = \begin{pmatrix} \hbar\omega + \sum_i V_x(\boldsymbol{R}_i) & t_{01} & t_{02} & t_{03} \\ t_{01} & V_e(\boldsymbol{R}_0) & 0 & 0 \\ t_{02} & 0 & V_e(\boldsymbol{R}_1) & 0 \\ t_{03} & 0 & 0 & V_e(\boldsymbol{R}_2) \end{pmatrix} \tag{44}$$

where:

- $\hbar\omega$ is the energy quantum of the system
- $V_x(\boldsymbol{R}_i) = a_{Vx}|\boldsymbol{R}_i|^2$ is the potential energy function
- $V_e(\boldsymbol{R}_i) = \sum_j V_x(\boldsymbol{R}_j) + V_a(\boldsymbol{R}_i) - V_x(\boldsymbol{R}_i)$ is the effective potential
- $V_a(\boldsymbol{R}_i) = a_{Va}(|\boldsymbol{R}_i - x_{\text{shift}}|^2 + c)$ is the additional potential
- t_{0i} are the transition dipole moments between states

The parameter vector $\boldsymbol{R}_\theta$ traces a perfect circle orthogonal to the $x = y = z$ line in 3D space, centered at the origin $(0, 0, 0)$, parameterized by angle θ.

5.7 Berry Connection and Berry Phase

The Berry connection $\boldsymbol{A}_n(\boldsymbol{R})$ for the n-th eigenstate is defined as:

$$\boldsymbol{A}_n(\boldsymbol{R}) = i\langle n(\boldsymbol{R})|\nabla_{\boldsymbol{R}}|n(\boldsymbol{R})\rangle \tag{45}$$

The Berry phase γ_n is the integral of the Berry connection around a closed loop C in parameter space:

$$\gamma_n = \oint_C \boldsymbol{A}_n(\boldsymbol{R}) \cdot d\boldsymbol{R} \tag{46}$$

5.8 Off-Diagonal Elements

The off-diagonal elements $\tau_{nm}(\theta)$ for $n \neq m$ provide information about the coupling between different eigenstates. These elements can be used to analyze transitions between states and to identify regions in parameter space where eigenstates are strongly coupled.

6 The Tau Matrix Method

We call this the "tau matrix method" because the key matrix is the tau matrix in the codebase. However, this matrix is actually the matrix of the Non-adiabatic coupling terms.

In this method, we first compute τ matrix, then use it to compute the Berry phase.

6.1 Definition of the Tau Matrix

This method is a numerical approach for computing the Berry phase. The key element is the tau matrix, whose elements are defined as:

$$\tau_{nm}(\theta) = \langle \psi_m(\theta) | \frac{\partial}{\partial \theta} | \psi_n(\theta) \rangle \tag{47}$$

where $|\psi_n(\theta)\rangle$ is the n-th eigenstate of the Hamiltonian $H(\boldsymbol{R}_\theta)$.

7 Numerical Results and Analysis

7.1 Berry Phase Matrix Structure

The numerical results show that the Berry phase matrix γ_{nm} has a specific structure with the following features:

- The diagonal elements γ_{nn} are typically small (order 10^{-3} to 10^{-4}), indicating minimal geometric phase accumulation for individual states.
- Large off-diagonal elements (order 10^0) appear in specific positions, particularly between states 1-2, 2-1, 1-3, and 3-1.
- The values close to $\pm 2\pi$ in the [1,2] and [2,1] positions indicate complete phase winding, which is a signature of non-trivial topology.

7.2 Physical Interpretation

The observed Berry phase structure has important physical implications:

- The system exhibits non-trivial geometric phase accumulation primarily in the subspace spanned by states 1 and 2.
- The antisymmetric nature of the off-diagonal elements ($\gamma_{12} \approx -\gamma_{21}$) is consistent with the expected behavior of a quantum system with time-reversal symmetry.
- The 2π phase difference indicates that the system's parameter space trajectory encloses a topological singularity.